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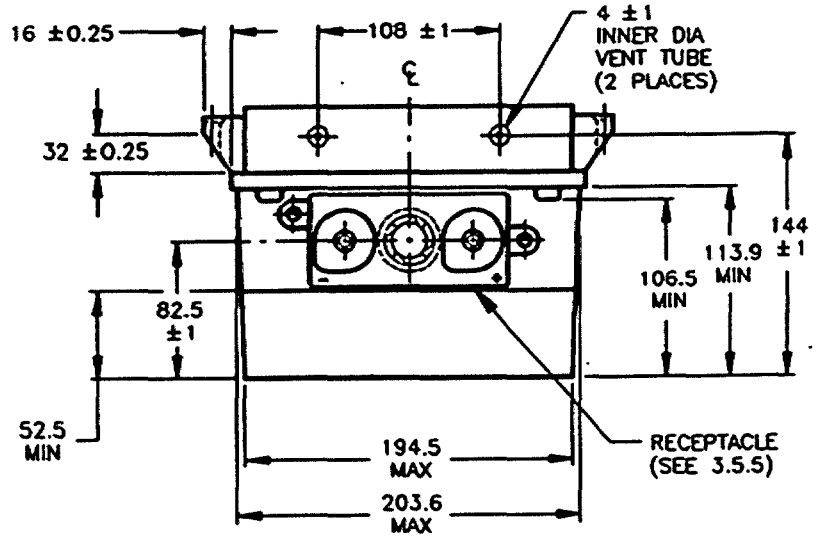
MIL-B-8565/8(AS)
19 November 1993

MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MEDIUM-RATE, TYPE 1,
MAINTENANCE-FREE, 24-VOLT, 15-AMPERE-HOUR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation:
MIL-B-8565



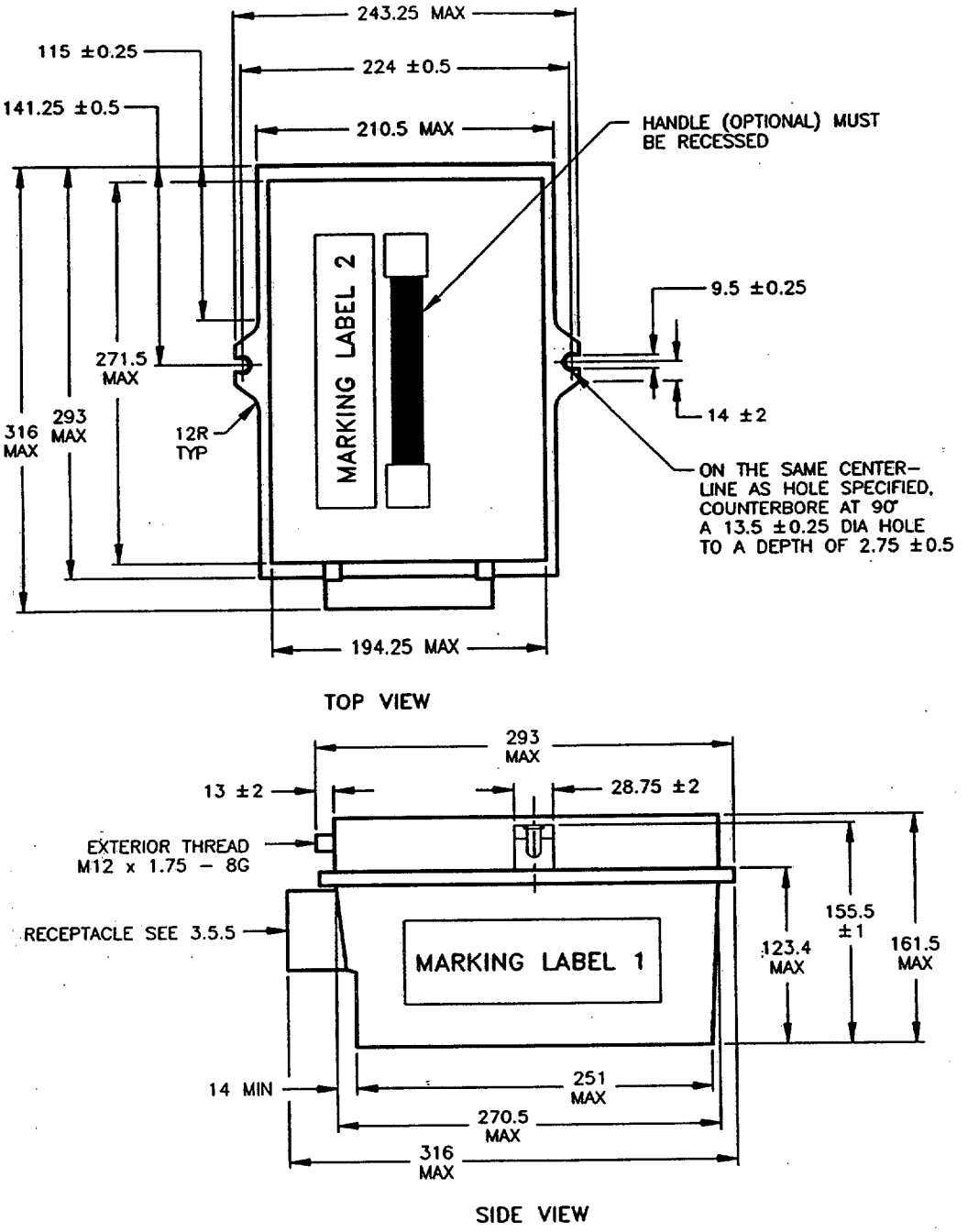
FRONT VIEW

NOTES:
1. All dimensions are in millimeters.
2. Abbreviations used in this figure are diameter (DIA), maximum (MAX), minimum (MIN) and part number (PN).

FIGURE 1. PN D8565/8-1 dimensions.

AMSC N/A FSC 6140
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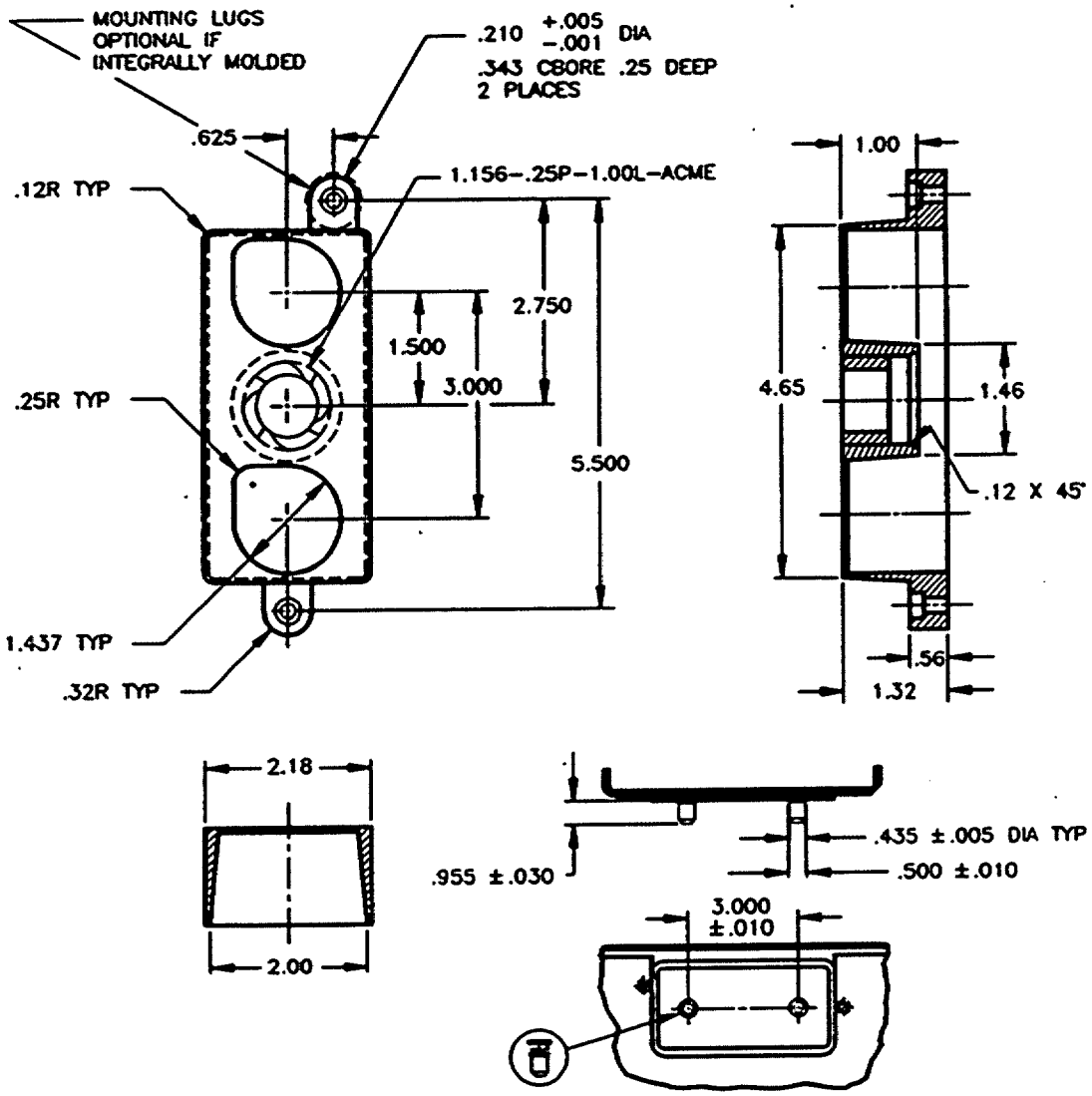
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NOTES:
1. All dimensions are in millimeters.
2. Abbreviations used in this figure are diameter (DIA), maximum (MAX), minimum (MIN), part number (PN), radius (R) and typical (TYP).

FIGURE 1. PN D8565/8-1 dimensions - Continued.

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NOTES:

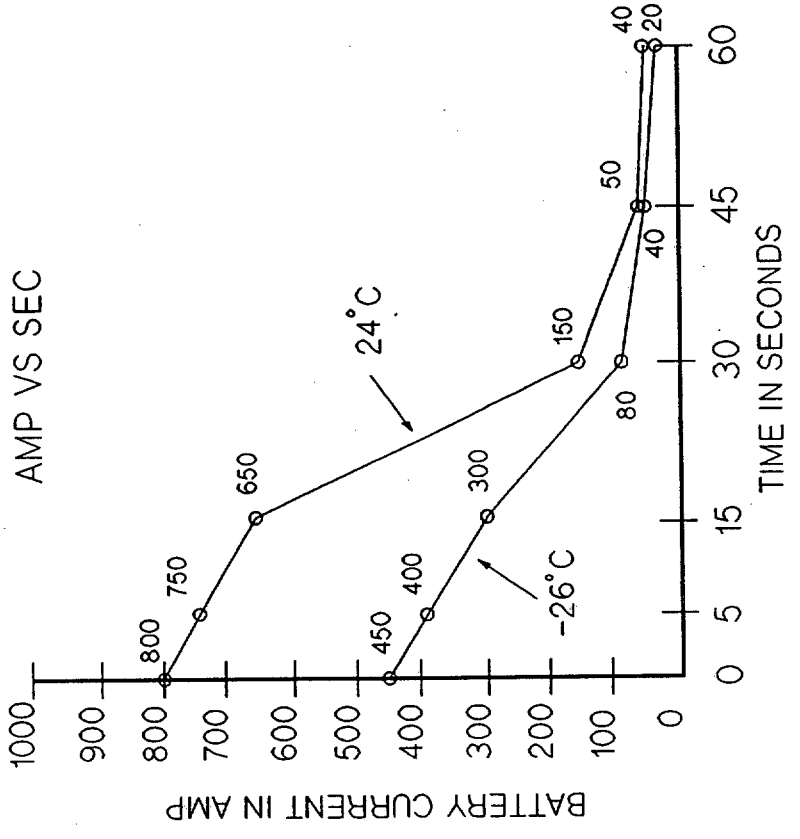
1. Material and finish to be acid resistant.
2. All dimensions are in inches.
3. Tolerances are X.XX (± .03 inches), X.XXX (± .010 inches).
4. Receptacle to case attachment method optional for nonmetallic containers.
5. Draft angles allowed in conformance with commercial die casting practice.
6. Abbreviations used in this figure are counterbore (CBORE), diameter (DIA), radius (R) and typical (TYP).

FIGURE 2. Typical receptacle adapter.

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MINIMUM DISCHARGE CURRENT VS TIME		
TIME IN SECONDS	CURRENT IN AMP	
	AT 24°C	AT -26°C
0+	800	450
5	750	400
15	650	300
30	150	80
45	50	40
60	40	20

AVERAGE DISCHARGE CURRENT DURING 14-V DISCHARGE SHALL BE NOT LESS THAN THE FOLLOWING AMP VALUES	
24°C	320.0 AMP
-26°C	165.0 AMP

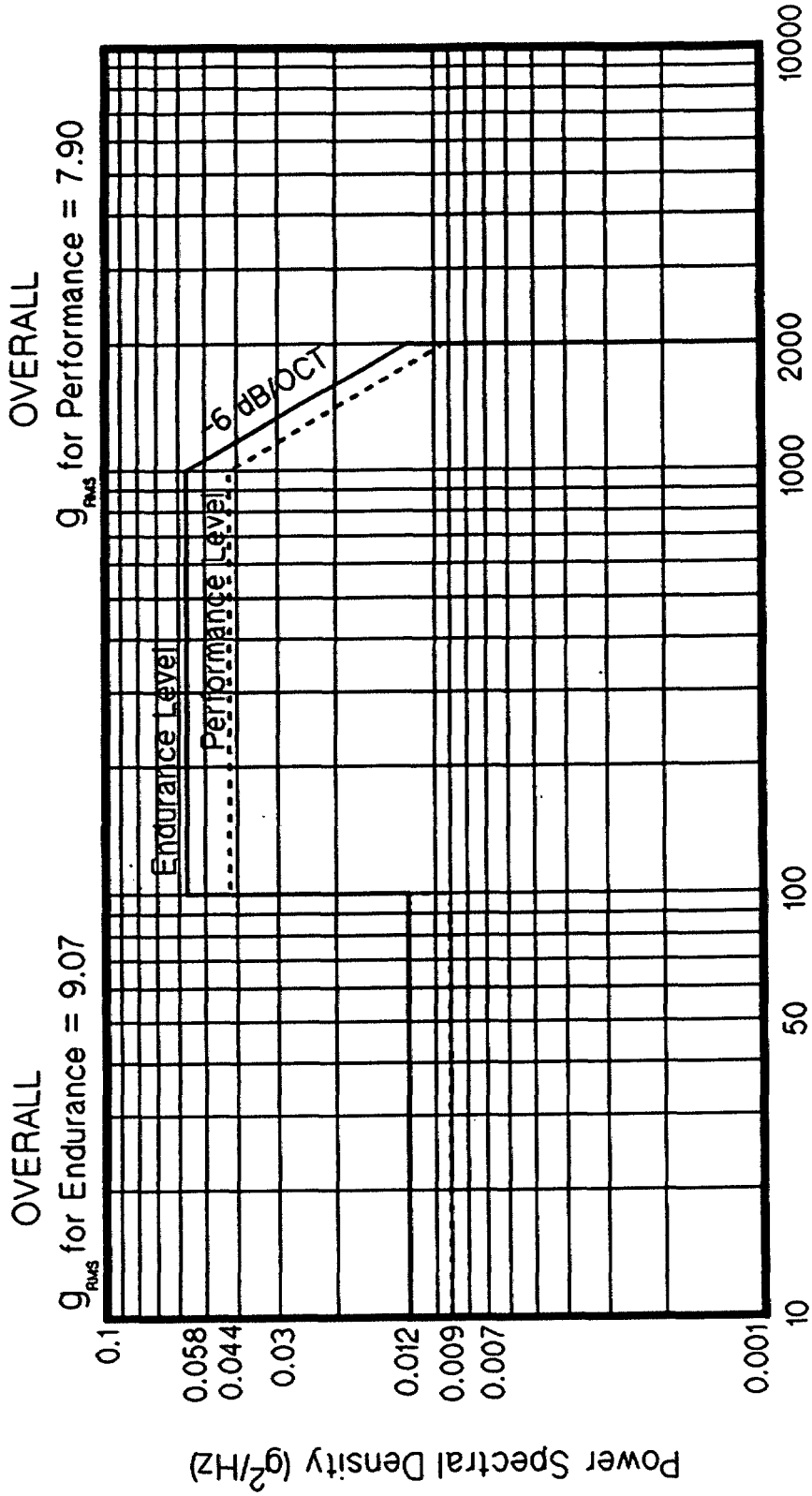


NOTE:

1. Abbreviations used in this figure are amperes (AMP), seconds (SEC), versus (VS), volts (M), and degree Celsius (°C).

FIGURE 3. Constant 14-V discharge curves.

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NOTE:
1. Abbreviations used in this figure are decibel (dB), hertz (Hz), acceleration of gravity (g), octave (OCT), and root mean square (RMS).

Figure 4. Random vibration test levels.

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REQUIREMENTS:

1. The dimensions of the battery shall be as shown on figure 1.

2. The part number of the battery shall be D8565/8-1.

3. The rated capacity of the battery shall be 15 ampere-hours based on its ability to conform to the life cycling final discharge (15.0 Ah/1 Hr/24°C/18.0V).

4. The weight of the battery shall be not greater than 19.5 kg.

5. Battery life in the aircraft shall be 3 years.

6. Table I lists the quality assurance tests.

7. The battery shall conform to MIL-B-8565 for type 1, medium rate, except as modified by the following.

3.5.4 Venting. Delete the following sentence: "Each vent tube shall be capable of supporting 1.5 times the battery weight."

3.5.5 Receptacles. Add the following: "The receptacle for the battery's positive and negative connections shall be dimensionally equal to figure 2 of the applicable specification sheet. The receptacle may be integrally molded in the container."

3.6.7 Vent tubes. Delete.

3.6.8 Capacity and electrical performance. Add the following: "During the 1-minute, 14-volt, constant voltage discharge, the battery shall meet at least one of the following conditions:

"Condition 1. The discharge current during the constant voltage discharge shall be not less than the ampere values shown on figure 3 of the applicable specification sheet. Inability of the battery to deliver the specified minimum discharge currents during the discharge shall constitute a requirement failure.

"Condition 2. The discharge current during the constant voltage discharge shall be not less than the ampere values shown on figure 3 of the applicable specification sheet throughout the first 15 seconds of the discharge. Furthermore, through the duration of the discharge, the battery shall maintain an average current not less than the average ampere value shown on figure 3 of the applicable specification sheet. Inability of the battery to deliver the specified currents during the discharge shall constitute a requirement failure."

4.6.7 Strength of vent tubes. Delete.

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4.6.23 Vibration. Delete and add the following: "Endurance and performance random test. The battery assembly, including connectors and cable, shall be mounted to the test apparatus in accordance with 4.5.5 and subjected to the random vibration test Procedure I, Category 5, Method 514.4 of MIL-STD-810 for equipment installed in jet aircraft. The battery shall be vibrated along each orthogonal axis in accordance with the random vibration test levels specified on figure 4 of the applicable specification sheet. The test sequence shall be as follows: 0.5 hours per axis at the performance level, 3 hours per axis at the endurance level and 0.5 hours per axis at the performance level for a total of 4 hours per axis. The battery shall be charged at 28.25 ± 0.5 volts during vibration and shall be discharged at 97.5 amperes for 2 minutes before beginning each axis of vibration. The cables connected to the battery shall consist of four size AN-6 cables (two connected to the positive terminal and two connected to the negative terminal) and shall be clamped to the test apparatus at a cable length of 46 ± 3 centimeters from the battery."

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TABLE I. Inspection of batteries and order of test.

Order of test	Description of inspection	Requirements paragraph	Inspection method paragraph	Qualification inspection			Quality conformance inspection		
				Qualification sample number			Group A tests	Group B tests	
				1	2	3		1	2
1	Visual & mechanical examination	3.5, 3.6.1; 3.6.3 & 3.6.22	4.6.3	X	X	X	X	X	X
2	Dimensions & weight	3.6.5	4.6.4	X	X	X	X	X	
3	Dielectric strength	3.6.4	4.6.5	X	X	X	X	X	X
4	Color & marking	3.6.3	4.6.8	X	X	X	X	X	X
5	Conditioning charge	4.5.1.4	4.6.9	X	X	X		X	X
6	Capacity discharge	3.6.8	4.6.10	X	X	X	X	X	X
7	Constant voltage discharge	3.6.8	4.6.11	X	X		X	X	
8	Strength of receptacle	3.6.9	4.6.12			X			X
9	Life cycling	3.6.10	4.6.13	X				X	
10	Capacity discharge at 49°C	3.6.8	4.6.14		X				
11	Charge & discharge test at low temperature	3.6.8	4.6.15		X				X
12	Discharge while inverted	3.5.8 & 3.6.8	4.6.16		X				X
13	Altitude	3.6.11 & 3.6.12	4.6.17		X				X
14	Mechanical shock	3.6.11 & 3.6.12	4.6.18	X					
15	Temperature shock	3.6.11 & 3.6.12	4.6.19	X				X	
16	Temperature rise & float	3.6.12 & 3.6.13	4.6.21		X				X
17	Battery gas emission test	3.6.11 & 3.6.14	4.6.22		X				
18	Vibration	3.6.11, 3.6.12 & 3.6.16	4.6.23		X				1/
19	Humidity	3.6.11, 3.6.12 & 3.6.17	4.6.24		X				X
20	Salt fog	3.6.11, 3.6.12 & 3.6.18	4.6.25	X				X	
21	Ground storage	3.6.15.1	4.6.26			X			
22	Physical integrity at high temperature	3.6.11 & 3.6.12	4.6.27	X					
23	Impact resistance test	3.6.19	4.6.27.1		2/				
24	Final examination	3.6.4	4.6.5 & 4.6.30	X	X	X		X	X

1/ Vibration will be performed randomly on lot samples.
2/ For nonmetallic battery container designs, two outer nonmetallic battery containers must be supplied. The containers shall be subjected to the impact resistance test of 4.6.27.

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